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Patterns and Sources of Personality Development in Old Age

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Abstract

Despite abundant evidence that personality development continues in adulthood, little is known about the patterns and sources of personality development in old age. We thus investigated mean-level trends and individual differences in change as well as the genetic and environmental sources of rank-order continuity and change in several personality traits (*neuroticism*, *extraversion*, *openness*, *agreeableness*, *conscientiousness*, *perceived control*, and *affect intensity*) and well-being. In addition, we analyzed the interrelation between *perceived control* and change in other personality traits as well as between change in personality traits and change in well-being. We analyzed data from older adult twins, aged 64-85 at time 1 (*N* = 410; 135 males and 275 females; 134 monozygotic and 63 dizygotic twin pairs), collected at 2 different time points about five years apart. On average, *neuroticism* increased, whereas *extraversion*, *conscientiousness*, and *perceived control* significantly decreased over time. Change in *perceived control* was associated with change in *neuroticism* and *conscientiousness* pointing to particular adaptation mechanisms specific to old age. Whereas individual differences in personality traits were fairly stable due to both genetic and environmental sources, individual differences in change were primarily due to environmental sources (beyond random error) indicating plasticity in old age. Even though the average level of well-being did not significantly change over time, individual well-being tended to decrease with strongly increasing levels of *neuroticism* as well as decreasing *extraversion*, *conscientiousness*, and *perceived control* indicating that personality traits predict well-being but not vice versa. We discuss implications for theory on personality development across the life span.

Keywords: Big Five traits; well-being; continuity; change; genetic; environmental

Patterns and Sources of Personality Continuity and Change in Old Age

A large number of empirical studies have examined the patterns of continuity and change in personality traits (see Roberts & Delvecchio, 2000, Roberts, Walton, & Viechtbauer, 2006, for meta-analyses) and their relation to well-being in adulthood (see Diener, Suh, Lucas, & Smith, 1999, for a review). They have shown average age trends and individual differences in change across the life span indicating that personality develops throughout the entire life.

In recent years, an increasing quantity of longitudinal behavioral genetic studies has addressed the genetic and environmental sources of personality development (see Bleidorn, Kandler, & Caspi, 2014, and Turkheimer, Pettersson, & Horn, 2014, for reviews). These studies suggest that continuity and change in personality differences are attributable to both genetic and environmental sources. The extent of continuity and change and the relative contribution of genetic and environmental factors, however, appear to depend on the age groups studied (Lucas & Donnellan, 2011; Kandler, 2012; Specht, Egloff, & Schmukle, 2011).

Although these studies have considerably advanced our understanding of adult personality development, less is known about the patterns and sources of personality development in old age (e.g., Specht et al., 2014). The current longitudinal twin study attempts to fill this gap by investigating the patterns and sources of continuity and change in personality traits as well as their interrelation with well-being in older adults.

Since the study of personality development requires an explicit theoretical model of personality differences, we first introduce the basic personality constructs of this study. We then review previous findings on personality development in adulthood and old age with a focus on mean-level trends, rank-order continuity, and the sources of personality development. Finally, we address the interrelation between personality and psychological well-being. In this regard, we illustrate theoretical perspectives which provide the basis for our hypotheses on patterns and sources of personality development in old age.

Personality Characteristics

Personality is most commonly defined as the sum or product of all characteristics that describe individual differences in thoughts, feelings, strivings, and behaviors that are relatively enduring across situations and over time within a certain reference population, such as age group or culture (Kandler, Zimmermann, & McAdams, 2014). There are several concepts, models, and theories on personality differences and development that are in line with this broad definition of personality (e.g., McAdams & Olson, 2010; McCrae & Costa, 2008; Shiner & DeYoung, 2013). While they disagree about the number of characteristics that are essential to describe individual differences in personality, they agree in the idea of a finite number of basic traits (or trait dimensions).

In this study, we focus on one basic model and two related concepts capturing personality differences. The basic and well-established model represents the Big Five taxonomy of trait dimensions (Digman, 1990; McCrae & John, 1992): (1) *neuroticism* vs. *emotional stability*, (2) *extraversion* vs. *introversion*, (3) *openness to experiences*, (4) *agreeableness*, and (5) *conscientiousness*. The Big Five model captures the trait landscape as portrayed in many different languages, societies, and cultures. It is useful to integrate various systems and inventories of personality description in a common framework (John, Naumann, & Soto, 2008).

Alternative concepts related to the construct personality focus on specific cognitive and affective features of psychological individuality. Two prominent and well-studied constructs are *locus of control* (Rotter, 1966, 1982) or *perceived control* (Skinner, 1994) and *emotional reactivity* or *affect intensity* (Larsen, Diener, & Emmons, 1986; Larsen & Diener, 1987). *Perceived control* refers to the extent to which individuals believe that they can control events they experience. It is conceptually and empirically linked to *self-esteem* and *generalized self-efficacy* (e.g., Judge, Erez, Bono, & Thoresen, 2002). *Affect intensity* is defined as the typical strength of individuals' emotional responsiveness in their everyday life and thus conceptually

and empirically linked to *emotional reactivity* and *emotion regulation.* It is distinct from the valence and frequency of affect (e.g., Schimmack & Diener, 1997), which are well-covered by *neuroticism* (i.e., negative emotionality) and *extraversion* (i.e., positive emotionality). Despite their questionable unidimensionality (Bryant, Yarnold, & Grimm, 1996; Coombs & Schroeder, 1988; Levenson et al., 1974; Weinfurt, Bryant, & Yarnold, 1994), *perceived control* and *affect intensity* can be captured in terms of generalized dimensional traits. *Perceived control* reflects a continuum from *externality* (i.e., life is controlled by environmental circumstances) to *internality* (i.e., life is under one's own control) *of control beliefs* (Rotter, 1990). *Affect intensity* captures a dimension from *low* to *high degree of strength in emotional reactivity* (Larsen, Billings, & Cutler, 1996).

Perceived control and *affect intensity* are distinct constructs but show systematic links to specific Big Five traits. *Perceived control* is negatively linked to *neuroticism* and positively linked to *extraversion* and *conscientiousness* (Judge et al., 2002; Zuckerman, Kuhlman, Joireman, Teta, & Kraft, 1993). *Affect intensity* is positively associated with *neuroticism*, *extraversion*, and *agreeableness* (Gohm & Clore, 2002). Generally, their associations with Big Five traits are only moderate and the changes in *affect intensity* and *perceived control* are expected to differ in old age (see below), requiring separate analyses of trait development.

Patterns of Personality Development

In the following, we review empirical findings on normative age-trends in personality traits and individual differences in personality change over time. Age trends reflect the average change (i.e., mean-level increase or decrease) on a single personality trait occurring in a population over time. However, individuals can differ in their level of change. This may alter the relative standing of individuals on personality traits compared to other individuals within a reference population (i.e., rank-order change) and (or) may lead to an increase or decrease of individual differences over time (without changing the rank-order of individuals).

Mean-Level Trends and Maturation

In their meta-analysis, Roberts et al. (2006) found that the average level of *neuroticism decreased*, whereas it increased for *agreeableness* and *conscientiousness.* This increase was more pronounced in young adulthood compared to older ages. Although the average level of *extraversion* remained relatively constant, they reported contrary trends for facets of extraversion (*social dominance* increased and *social vitality* decreased). *Openness* tended to show a curvilinear pattern with increases in young adulthood and decreases in later adulthood. This pattern of mean-level trends showing decreases in *neuroticism* and increases in both *conscientiousness* and *agreeableness* has sometimes been labeled social maturation because increases in these traits are important for successful social integration and functional for mastering developmental tasks (*maturity principle*; Roberts & Wood, 2006). However, information on older adults was limited in this meta-analysis. So the question remains to what extent personality maturity continues up to the highest ages.

Recent studies using data from large representative samples mainly confirm the previous results on age trends in young and middle adulthood (e.g., Lucas & Donnellan, 2011; Soto, John, Gosling, & Potter, 2011; Specht et al., 2011; Wortman, Lucas, & Donnellan, 2012), but yield heterogeneous results for later adulthood. In a study of older adults, Mõttus, Johnson, and Deary (2012) found small or no significant mean-level changes for Big Five traits in a cohort between the ages 69 and 72, but reported that an older cohort showed significant decreases in *extraversion*, *intellect* (i.e., *openness*), *agreeableness*, and *conscientiousness*, whereas *neuroticism* increased (at least for women) between the ages 81 and 87. Similar results were found in a study using a representative Australian sample (Wortman et al., 2012). Whereas *neuroticism* tended to decrease and *agreeableness* as well as *conscientiousness* tended to increase from young to middle adulthood, they showed opposite trends at the latest after age

80. These studies suggest that the strength of mean-level change in old age is comparable to those under age 30, even though the trend goes into the opposite direction.

Also in line with the *maturity principle*, *perceived control* increases with age, at least from younger ages to middle adulthood (Heckhausen, Wrosch, & Schulz, 2010; Lachman, Neupert, & Agrigoroaei, 2011). It may not surprise that reverse trends were reported for older ages (e.g., Specht, Egloff, & Schmukle, 2013a) because older adults see fewer opportunities for control due to increasing acknowledgement of physical and cognitive constraints and limitations (Lachman et al., 2011; Wrosch, Heckhausen, & Lachman, 2006).

The average trend of change for *affect intensity* (or *emotional reactivity*) in old age is less clear. Whereas some studies suggest that it declines with age from younger to later adulthood (Diener, Sandvik, & Larsen, 1985; Kandler, Riemann, & Angleitner, 2013) probably due to enhanced capabilities of affect regulation with age, others have reported no changes or even increases in emotional reactivity due to losses of the ability to regulate emotions in older ages (see Kessler & Staudinger, 2009; Kunzmann, Richter, & Schmukle, 2013; Scheibe & Carstensen, 2010).

In the first instance, the findings of reverse trends for personality development in old age seem to contradict the *maturity principle* as formulated by Roberts, Wood, and Caspi (2008): “People become more socially dominant, agreeable, conscientious, and emotionally stable with age” (p. 376). However, in terms of successful social integration and functionality for mastering developmental tasks (Roberts & Wood, 2006), reverse trends in old age may not contradict the *maturity principle* per se because they may reflect normative adaptations to possible or anticipated deterioration of health status and cognitive functionality (Baltes, 1997). Therefore, we propose a more fine-grained formulation of maturation for different age groups in terms of life-management strategies of selection, optimization, and compensation (SOC; Baltes, Lindenberger, & Staudinger, 2006; Freund & Baltes, 1998, 2002).

The SOC perspective provides the basis for explanations of different adaptive mechanisms in old age compared to younger ages due to a different ratio of gains to losses: Older people have to be more wary of health-risks and dangers of everyday life (i.e., increasing *neuroticism*), more selective in their activities and investments in social relationships (decreasing *extraversion* and *agreeableness*), cultural and intellectual life (decreasing *openness to experiences*), and effort they can put into being orderly and dutiful (decreasing *conscientiousness*) when their physical, social, and cognitive functionality (i.e., *perceived control*) declines to retain a comparable level of emotion regulation (stable *affect intensity*) and well-being (Brandtstädter & Rothermund, 1994; Mõttus, Johnson, Starr, & Deary, 2012). This hypothesis also suggests that change in *perceived control* may predict change in Big Five personality traits and it is in line with findings on rather stable average levels of well-being in later life (Carstensen, Pasupathi, Mayr & Nesselroade, 2000; Kunzmann, Little, & Smith, 2000).

Rank-Order Continuity and Individual Differences in Change

Despite expectable normative trends in personality traits for older adults, not all individuals change alike, resulting in different individual trajectories and non-perfect rank-order continuity (i.e., the degree to which individual differences are stable over time). Meta-analyses of longitudinal studies and cross-sequential studies using large representative samples consistently suggest an inverted U-shaped function of rank-order continuities (for equal intervals of time) across adulthood (Ardelt, 2000; Lucas & Donnellan, 2011; Roberts & DelVecchio, 2000; Specht et al., 2011, 2013a; Wortman et al., 2012). Rank-order continuity increases from childhood to middle adulthood and peaks around the age of 50. This pattern of increasing continuity has been referred to as the *cumulative continuity principle* (Caspi, Roberts, & Shiner, 2005). After age 60, however, rank-order continuity begins to cumulatively decrease again. That is, similar to enhanced mean-level change, individual differences in trajectories appear to be as strong in old age as is in young adulthood.

The increasing variance in individual change in later adulthood may not be surprising, because people differ in the beginning and the strength of deterioration of health status and cognitive functionality in old age (Freund & Baltes, 1998, 2002). Even though deterioration of physiological and psychological functionality is an inevitably progressing biological process (Baltes, 1997), environmental factors can accelerate (e.g., due to drugs, stress, bad nutrition) or decelerate (e.g., due to healthy food, proper physical activity) this process. Furthermore, the number of life events and transitions (e.g., retirement, widowhood, death of peers, care giving duties) increases (Plomin, Lichtenstein, Pedersen, McClearn, & Nesselroade, 1990) and the stabilization in mid-life, which is due to stable social roles (e.g., in family and job), decreases in old age (Freund, Nikitin, & Ritter, 2009). Consequently, large individual differences in personality change in old age may be due to environmental sources. This is in line with the *plasticity principle* of personality development (Roberts et al., 2008): "Personality traits are open systems that can be influenced by the environment at any age" (p. 376).

Sources of Personality Development

In the last two decades, behavioral genetic studies have provided important insights into the genetic and environmental sources of continuity and change in personality differences over time and ages. They have shown that both genetic and environmental factors are driving forces of continuity as well as change (Bleidorn et al., 2014). Despite the fact that genetic and environmental factors do not independently influence personality development but correlate and interact in complex ways, the relative contributions of these sources to continuity and change in individual differences vary across different age groups. This has important implications for theory in personality development (Specht et al., 2014).

From childhood to young adulthood, genetic factors represent the major source of continuity in personality differences, whereas both genetic and environmental factors contribute to individual differences in personality change (Bratko & Butkovic, 2007; De Fruyt et

al., 2006; McGue, Bacon, & Lykken, 1993; Spengler, Gottschling, & Spinath, 2012). The latter indicates that maturation is an individual process in which biological and environmental factors are interwoven. However, the relative contribution of genetic factors to rank-order changes appears to decrease with age, indicating decreasing genome-based plasticity (Baltes, 1997), whereas environmental sources of rank-order continuity appear to increase from adolescence to young adulthood (Gillespie, Evans, Wright, & Martin, 2004; Kandler, 2012; Hopwood et al., 2011). Genetic factors reach almost perfect stability at the beginning of middle adulthood indicating that individual differences in genetically induced maturation are negligible beyond the age of 30 (McCrae & Costa, 2008). Environmental influences, however, seem to increasingly stabilize personality differences throughout adulthood (Bleidorn, Kandler, Riemann, Angleitner, & Spinath, 2009; Johnson, McGue, & Krueger, 2005; Kandler et al., 2010; Pedersen & Reynolds, 1998; Viken, Rose, Kaprio, & Koskenvuo, 1994). This may be attributable to stabilizing social, living, and working contexts in middle adulthood. That is, social roles individuals engage in different contexts (*social investment principle*; Roberts & Wood, 2006) may be largely stable in middle adulthood contributing to high environmental continuity in personality differences (*role continuity principle*; Roberts et al., 2008).

Less is known about the relative contributions of genetic and environmental factors to continuity in personality differences and personality change in old age. The only existing study with data from age groups with a mean age beyond 70 exclusively focused on *neuroticism* and *extraversion* (Read, Vogler, Pedersen, & Johansson, 2006). The results indicated that genetic factors remained perfectly stable contributing to continuity in personality differences, whereas environmental factors largely contributed to individual differences in personality change. That is, the decrease in rank-order continuity in older age appears to be due to environmental factors. This finding may not surprise, because many new environmental (e.g., role) demands in the face of developmental transitions (e.g., retirement), events (e.g., loss of partner by death),

and life circumstances (e.g., own chronic disease) occur in old age that require new adaptations (e.g., managing caregiving duties, handling loneliness and deterioration in health). This is in line with the *plasticity principle* of personality development.

Read et al. (2006) provided no evidence of genetic influences on individual differences in personality change in old age, indicating less relevance of genetically driven individual differences in the process of deterioration of physiological functionality for personality change. Recent phenotypic research also suggests that physiological health declines have only small influences on changes in personality in old age (Berg & Johansson, 2014; Magee, Heaven, & Miller, 2013; Mõttus, Johnson, Starr, & Deary, 2012; Mroczek & Spiro, 2003; Takahashi, Edmonds, Jackson, & Roberts, 2013). Because of low associations between physiological and psychological health (Okun & George, 1984), change in subjective well-being (i.e., people's judgments and moods about their health and life quality) rather than change in objective health may be linked to personality change. However, the paucity of studies on old age groups leaves open questions with regard to this issue.

The Interrelation between Personality Traits and Psychological Well-Being

Whereas most studies have found rather stable average levels of well-being in later life (e.g., Carstensen et al., 2000; Diener et al., 1999; Kunzmann et al., 2000), others suggest a terminal decline when people move closer to death (Gerstorf et al., 2010). That is, although older people manage to retain a comparable level of psychological well-being in general, well-being may decline for some people and this change in well-being may be associated with changes in personality traits. Only a handful of studies have examined the interdependence of personality and well-being change. Previous research suggests that more emotionally stable, extraverted, agreeable, and conscientious individuals experience higher levels of well-being (see Steel, Schmidt, & Shultz, 2008, for a meta-analysis), and that changes in these traits correlate with changes in well-being (e.g., Boyce, Wood, & Powdthavee, 2013). Whereas some

studies have reported that personality traits influence well-being (Charles, Reynolds, & Gatz, 2001; Griffin, Mroczek, & Spiro, 2006), more recent studies suggest support for a reciprocal relationship between personality traits and well-being (Soto, 2014; Specht, Egloff, & Schmukle, 2013b). The latter can be interpreted in terms of two pathways: (1) Personality traits represent the relatively stable and genetically anchored core of subjective well-being (Headey, 2006; Headey & Wearing, 1989), but (2) changes in states of well-being due to environmental fluctuations may, in turn, affect changes in personality traits (Specht et al., 2013b; van Aken, Denissen, Branje, Dubas, & Goossens, 2006).

Studies which investigated the interrelation between personality traits and subjective well-being, however, have usually considered and measured subjective well-being as a trait-like concept in terms of global self-assessments about the quality of life. Those assessments usually show levels of rank-order continuity and heritability (i.e., the degree to which relative differences among individuals are due to genetic factors) comparable to personality traits (Bartels & Boomsma, 2009; Eid & Diener, 2004). Thus, subjective well-being itself may reflect a personality characteristic to some degree (Kandler et al., 2014). This raises the question whether studying the interrelation between personality traits and well-being is reasonable. For example, given the relative high correlation between *neuroticism* and facets of subjective well-being (e.g., happiness, life satisfaction, and quality of life; Steel et al., 2008), it is unclear what kind of interrelation is investigated: An association (1) between different personality traits; (2) between different aspects of the same personality construct; or (3) in fact between personality and well-being. Because of these considerations and since global assessments of life satisfaction or happiness did not show strong associations with aggregated state-measures of well-being (Thomas & Diener, 1990), psychological well-being should be captured as judgments and moods about momentary states and current living conditions (Bradburn, 1969) rather than trait-like self-related schemata.

Aims of the Current Study

The current longitudinal twin study was designed to investigate patterns and sources of continuity and change in personality traits (*neuroticism*, *extraversion*, *openness*, *agreeableness*, *conscientiousness*, *perceived control*, and *affect intensity*) in old age (age range: 64-89) over two measurement occasions about five years apart. In addition, we studied the relations between personality development and change in psychological well-being. We had five specific aims.

First, we examined mean-level trends in personality traits and well-being. According to the hypothesis of different adaptive mechanisms for different age groups in terms of maturation and life management that was presented above, we expected reverse trends for older compared to younger and middle aged persons (Hypothesis 1a): an increase in *neuroticism* and decreases in *extraversion*, *openness*, *agreeableness*, *conscientiousness*, and *perceived control*. Although the valence of emotionality should change (i.e., increasing negative emotionality or *neuroticism* and decreasing positive emotionality or *extraversion*), the average level of emotional reactivity (i.e., *affect intensity*) and well-being should not change (Hypothesis 1b) because of a balance between losses of emotion regulation ability and enhanced expertise in management strategies of affect regulation in older ages.

Second, our refined formulation of the maturity principle for older ages also suggests that increases in *neuroticism* and decreases in *extraversion*, *openness*, *agreeableness*, and *conscientiousness* act to compensate decreases in *perceived control*. Therefore, we also examined the interrelation between *perceived control* and Big Five personality traits. The longitudinal study design allows us to test four types of associations between two variables: initial level correlations, prospective effects in the one direction, prospective effects in the reverse direction, and change correlations. According to our hypothesis, we expected that *perceived control* prospectively predicts Big Five personality traits and that change in Big Five traits is associated with change in *perceived control* (Hypothesis 2).

Third, we investigated the degree of rank-order continuity and individual differences in change in personality traits and well-being. Beyond the fact that personality traits should be rather stable and more stable than well-being (Hypothesis 3a), we hypothesized in line with the *plasticity principle* of personality development that individual ranks in personality traits significantly change beyond random fluctuations due to measurement error (Hypothesis 3b).

Fourth, we disentangled genetic from environmental components of rank-order continuity and change in personality traits and well-being. As previous studies on *neuroticism* and *extraversion* suggested (Read et al., 2006; see also Kandler, 2012), we hypothesized that rank-order continuity in all personality traits should be due to genetic and environmental factors (Hypothesis 4a). In line with the *plasticity principle*, however, change in old age should be attributable to environmental factors (Hypothesis 4b). In particular, rather momentary assessments of well-being should depend on environmental circumstances.

Finally, we investigated the interrelation between personality traits and psychological well-being in old age. Consistent with previous research (e.g., Boyce et al., 2013; Steel et al., 2008), we expected that individual differences in levels and changes in well-being are significantly associated with individual differences in levels and changes in *neuroticism*, *extraversion*, *conscientiousness*, *agreeableness*, and *perceived control* (Hypothesis 5). As reviewed above, results from recent studies indicate a reciprocal relationship between personality traits and well-being (e.g., Soto, 2014; Specht et al., 2013b). That is, prospective trait effects on well-being might be as strong as prospective well-being effects on traits. However, as outlined above, most previous studies used trait-like measures of subjective well-being that might have led to artificial reciprocity between different traits or between different aspects of the same personality construct. Therefore, we captured psychological well-being as judgments of and feelings on the quality of their current living conditions in old age to test the reciprocity of effects between personality traits and well-being.

Method

Participants

The data used in this study were drawn from the Genetic Oriented Lifespan Study on Differential Development (GOLD). This twin study was initiated in 1937 with 90 twin pairs between six and 18 years of age. For a follow-up study (between 1994 and 1999) with a focus on later adulthood only 20 complete twin pairs of the initial sample could be tested. Therefore, the sample was extended by new pairs of 64- to 85-year old twins (Geppert & Halisch, 2001). The upper age cut was chosen to reduce the probability of dropout due to mortality. After an interval of almost five years (*M* = 4.5), the participants were re-tested (Halisch & Geppert, 2012). These last two follow-up waves contained an adequate sample size and several new personality questionnaires and measures of subjective well-being which captured the constructs of interest for the current study. Therefore, only data from the last two measurement waves (named Time 1 and Time 2 in the following) were analyzed in this study.

The total sample at Time 1 consisted of 410 individuals (135 males and 275 females) including 134 complete MZ and 63 complete DZ twin pairs (91 female and 43 male MZ and 42 female and 21 male DZ pairs). The average age was *M* = 71.32 (range: 64-85). At Time 2, only 324 (79%) individuals (115 males and 209 females) including 148 complete twin pairs (101 MZ and 47 DZ pairs) participated. About 17% of the dropout sample died between Time 1 and Time 2. The average age was *M* = 75.63 (range: 68-89) at Time 2. Although the dropout tended to be larger for older people and women, MCAR (missing-completely-at-random) tests (Little, 1988) indicated that dropout could be assumed to be randomly distributed.

Measures

Personality traits. The 60-item German Neuroticism-Extraversion-Openness-Five-Factor-Inventory (NEO-FFI; Borkenau & Ostendorf, 1993; Costa & McCrae, 1992) was used to capture Big Five trait scores on *neuroticism*, *extraversion*, *openness*, *agreeableness*, and

conscientiousness. Each NEO-FFI item is a trait-descriptive self-assessment which participants rated on a 5-point scale from 1 = “does not describe me at all” to 5 = “describes me very well”. *Perceived control* was assessed using a 32-item questionnaire on generalized competence and control beliefs regarding social and health issues (Krampen, 1991). Twelve items were positively formulated (e.g., “I can determine a lot of things which happen in my life”) with respect to control and 20 items were reversely coded (e.g., “I often don’t know how I can achieve my wishes”). Each item had to be rated on a 6-point scale from 1 = “don’t agree” to 6 = “absolutely agree”. *Affect intensity* was captured by a German version of the 40-item Affect Intensity Measure (AIM; Larsen & Diener, 1987). The AIM examines emotional reactions to typical life circumstances rated on a 6-point scale from 1 = “never” to 6 = “almost always”.

Well-being. Psychological well-being as satisfaction with and moods about momentary states and current living conditions was measured with eight items drawn from Bradburn’s (1969) Affect Balance Scale. The items started with the question: “Did it happen in the last weeks or months that…” followed by four positive states (e.g., “…you were excited about something”) and four negative moods (“…felt unhappy.”). The participants had to answer “Yes” or “No” to each question. The negative states and moods were negatively coded.

Latent variable scores. To use personality and well-being latent variable scores corrected for measurement error, the items on each scale were assigned to two parcels as quasi-parallel tests. The split-half reliabilities of trait and well-being scores based on these parcels and internal consistencies of parcels for Time 1 and Time 2 as well as rank-order continuity for parcels, latent variables, and residuals are presented in Table 1 (*M*s and *SD*s parcel scores and true score correlations are shown in the supplementary Tables S1 and S2).

Analyses

Latent change models. To estimate mean-level trends and individual differences in trajectories for all variables, we ran latent change model analyses (Steyer, Eid, &

Schwenkmezger, 1997; McArdle, 2009). These structural equation models (SEMs; see Figure 1) disentangle latent variable scores at Time 1 and Time 2 into level and change scores. Developmental trajectories for the variables were estimated using these two latent variables. The level (or intercept) variable represents individual differences in the variable that remain stable across assessment waves, whereas the change (or slope) variable reflects individual differences in linear change over time. To interpret the latent change as true change the level and change variables have to be correlated. Given no change in the amount of variance across time, these correlations have a negative sign. Since these correlations depend on the direction and the degree of change as well as on the degree of the increase and decrease of variance across time, they should be (if at all) interpreted with caution and, thus, we avoided the interpretation of correlations between level and change.

Latent change twin models. Latent change SEM analyses are informative about the patterns of developmental trajectories but not about the sources. To estimate the genetic and environmental influences on individual differences in level and change, we ran latent change twin model analyses (Bleidorn et al., 2009; Kandler et al., 2013). These twin models (see Figure 2) allow the disentanglement of additive genetic factors from environmental factors shared and not shared by twins controlled for non-random method (due to parcel-specific biases) and random error variance.

Structural equation twin modeling (see Neale & Maes, 2004) is based on the difference in the proportion of segregating genes shared between MZ twins (100%) and DZ twins (on average 50%) and on the assumptions that components of environmental influences are equal for MZ and DZ twins, the absence of assortative mating of twins' parents with respect to the traits of interest, and no interplay between genetic and environmental factors. Larger MZ compared to DZ twin correlations are thus attributable to differences in the genetic similarity of MZ and DZ twins indicating genetic influences. The model also assumes the absence of

nonadditive genetic effects, because in a model of twins reared together these effects cannot be detected in the presence of shared environmental effects. Most of the assumptions are unproblematic with respect to personality characteristics, such as the equal environment assumption (Borkenau, Riemann, Angleitner, & Spinath, 2002) and the absence of assortative mating (Watson et al., 2004). In addition, estimates of additive genetic effects derived from twin models are quite good estimations of broad-sense heritability including additive and nonadditive genetic factors (Hahn et al., 2012; Hill, Goddard, & Visscher, 2008).

Latent crossed-lagged regression models. The aforementioned latent change SEM analyses provide patterns and estimates of the sources of continuity and change, but they are not informative about the directionality of effects between two variables (e.g., between *perceived control* and Big Five personality traits). To address this issue, we ran a series of bivariate latent autoregressive models known as crossed-lagged regression models (McArdle, 2009). These models (see Figure 3) allow estimations of prospective effects from one variable to another and vice versa controlled for initial correlations between the variables and their specific rank-order continuity. In addition, the models allow estimations of change correlations.

Statistical software and estimation procedures for SEM. We fitted all models using AMOS 21 (Arbuckle, 2012). To handle missing values due to the random dropout and to reach maximum statistical power the models were fitted to all available data via full information maximum likelihood (FIML) procedures (Little & Rubin, 2002). The significance of SEM parameters was evaluated by critical ratios (C.R. = estimates/standard error).

Results

Patterns of Continuity and Change

As shown in Table 1, rank-order continuity for latent true scores was quite high for personality traits (> .80) except for *affect intensity*. That is, individual differences in personality traits are largely but not perfectly stable over time in old age pointing to individual differences

in change in this period of life. Individual differences in well-being captured as rather momentary moods and judgments about current living conditions showed moderate continuity.

The results of latent change model analyses are shown in Table 2. The first columns refer to the latent level variable. The mean of the level reflects the average score at Time 1 and the variance represents the amount of individual differences at Time 1. Age was significantly correlated with the level in *extraversion*, *conscientiousness* and *perceived control*. The correlations suggest that older people perceive lower levels of control and tended to be less extraverted and conscientious. Sex was significantly related to the level of nearly all variables except *affect intensity* and *openness*. These associations suggest that women score higher in *neuroticism* and *agreeableness*, whereas men score higher in *extraversion*, *conscientiousness*, *perceived control*, and *well-being*. The correlations with sex are in line with previous studies (e.g., Costa, Terracciano, & McCrae, 2001; Diener et al., 1999; Lachman et al., 2011).

Latent change means indicated significant average changes over time for *neuroticism*, *extraversion*, *conscientiousness*, and *perceived control*. *Neuroticism* increased while *extraversion*, *conscientiousness*, and *perceived control* decreased across time. Standardized average changes (see Figure 4) also indicate that *openness* and *agreeableness* showed declines by trend. For all variables, variance in latent change scores were significant pointing to substantial individual differences in change. However, individual differences in change were neither significantly related to age (except for *extraversion*) nor to sex differences.

Sources of Continuity and Change

The proportions to which individual differences in level and change were due to genetic, shared environmental, and nonshared environmental influences are shown in Table 3. Level variance in all personality traits was attributable to both significant genetic and nonshared environmental influences. For *agreeableness*, model fitting analyses additionally yielded significant shared environmental influences. On average, genetic factors reflected the primary

source of rank-order continuity in personality traits (59%), whereas environmental factors contributing to continuity in personality traits were primarily not shared (30%). In our sample of older twins, shared environmental influences reflected the primary source of continuity in *agreeableness* and nonshared environmental influences represented the primary source of continuity in *affect intensity*.

For latent well-being scores, model fitting analyses yielded significant level components neither due to genetic factors nor attributable to shared environmental influences. That is, environmental influences unique to individuals represented the primary source of rank-order continuity in well-being.

Regarding individual differences in personality trait change, model fitting results again suggest significant genetic and environmental factors. On average, nonshared environmental influences reflected the primary source of change (53%), whereas shared environmental influences on the variance in individual change were negligible (4%). Even though genetic factors explained on average about 43% of the variance in change, only the genetic variance components in *extraversion* and *conscientiousness* were significant, whereas change in all personality traits (except *extraversion*) was significantly attributable to nonshared environmental influences.

Model fitting results also suggest genetic contributions to change in *openness*, *agreeableness*, and *perceived control* as large in size as the nonshared environmental component of change, even though the genetic variance components were not statistically significant. The statistical insignificance of the genetic component compared to the nonshared environmental component was due to the problem that distinguishing between additive genetic effects and shared environmental effects as factors accounting for MZ twins' resemblance needs more statistical power than detecting nonshared environmental factors which represent the only source accounting for MZ twins' dissimilarity. Rerunning the model

analyses for *openness*, *agreeableness*, and *perceived control* after fixing the shared environmental component of change (c_C^2) to zero did not lead to significantly worse model fits (Δ-2*LL* < 0.01; *df* = 1; *p* > .05) but to significant estimates of genetic influences on individual differences in change for these variables (shown in italics in Table 3).

Findings on the sources of change in well-being largely paralleled the results for continuity in well-being. Nonshared environmental influences represented the primary source, whereas genetic and shared environmental components were small and not significant.

The Interrelation between Perceived Control and Big Five Personality Traits

To investigate whether increases in Big Five traits might compensate decreases in *perceived control*, we ran five crossed-lagged regression model analyses. The results are presented in Table 4. In line with previous cross-sectional research (e.g., Judge et al., 2002; Zuckerman, 1993), all initial correlations were significant and suggest highest correlations with *neuroticism*, *extraversion*, and *conscientiousness*. The within regression weights indicated that these links are largely stable across time (β_T > .84 and β_{PC} > .77).

We did not find significant prospective effects from Big Five traits on *perceived control* or vice versa, except for *perceived control* on *conscientiousness* indicating that higher *perceived control* at Time 1 predicted lower levels in *conscientiousness* at Time 2. In the first instance, this seems to be counterintuitive given the substantial initial positive links between these traits. However, change in *conscientiousness* was also substantially and positively linked to change in *perceived control* and on average both variables showed significant declines over time (see Table 2 and Figure 4). Thus, the small but significant negative prospective effect from *perceived control* on *conscientiousness* simply reflected the fact that high initial levels of *perceived control* predicted subsequent stronger decreases in conscientiousness.

The analyses also yielded significant negative change correlations between *neuroticism* and *perceived control*. Increasing *neuroticism* was associated with decreasing *perceived control*

and vice versa. Although the change correlations between *perceived control* and the other Big Five traits were not significant, they point into the expected direction. That is, decreases in *perceived control* may be linked to decreases in *extraversion*, *openness*, and *agreeableness*.

The Interrelation between Personality Traits and Well-Being

Finally, we investigated the interrelation between personality traits and well-being in old age. The findings of the crossed-lagged regression analyses are shown in Table 5. Consistent with previous cross-sectional studies (e.g., Steel et al., 2008; Judge et al., 2002), higher initial well-being levels were substantially associated with lower levels of *neuroticism* and higher scores on *extraversion*, *conscientiousness*, and *perceived control*.

Whereas well-being did not show significant prospective effects on personality traits, the effects of initial personality traits on subsequent well-being largely paralleled the initial correlations indicating that personality trait levels affect well-being but not vice versa (except by trend for *perceived control*). That is, low initial levels of *neuroticism* and high initial levels of *perceived control*, *conscientiousness*, and *extraversion* (by trend) predicted higher levels of well-being at time 2. As a consequence, a substantial proportion of the differential stability in well-being could be accounted for by the rank-order continuity in *neuroticism*, *extraversion*, *conscientiousness*, and *perceived control* (compare $\beta_{WB} \leq .40$ in Table 5 with true score rank-order continuity of well-being in Table 1: .44).

Change correlations between personality traits and well-being also mirrored the initial correlations to some degree. That is, individuals who showed stronger decreases in *neuroticism* and increases in *perceived control* and *conscientiousness* (by trend) had a higher probability to show increases in well-being (despite no average trend in well-being, see Table 2 and Figure 4).

Discussion

The current study aimed to extend our knowledge about the patterns and sources of personality development and the interrelation between personality traits and well-being in old

age. The findings largely support our hypotheses and have different implications for maturation and plasticity of personality traits in later adulthood.

Average Personality Trends in Old Age

Consistent with our expectation (i.e., Hypothesis 1a: specific adaptive mechanisms for older adults in terms of different life-management strategies compared to younger adults) and in line with recent findings (Mõttus, Johnson, and Deary; 2012; Wortman et al., 2012), our results suggest that mean-level change in personality traits in old age drifts in the opposite direction as for younger adults (Roberts et al., 2006). Older people become aware of losses in cognitive and physical functionality (i.e., reduced *perceived control*) and, thus, have to be more wary of health-risks and dangers of everyday life (i.e., increasing *neuroticism*), more selective in their activities and investments in social relationships (i.e., decreasing *extraversion* and *agreeableness*), cultural life, intellectual challenges (i.e., reduced *openness to experiences*), and effort they can put into being orderly and dutiful (i.e., decreasing *conscientiousness*). In line with Hypothesis 1b, the results indicate that *affect intensity* and well-being tend to remain constant in later adulthood. This may reflect a positive balance between gains and losses in terms of successful mastery (i.e., optimized emotional and social functioning with maturity) of losses (i.e., reduced ability to regulate emotions and to invest in social interactions).

The aforementioned adaptive mechanisms also suggest that increasing *neuroticism* and decreasing *extraversion*, *openness*, *agreeableness*, and *conscientiousness* may act to compensate perceived losses in social and cognitive functionality (Hypothesis 2). We examined this hypothesis by analyzing the directionality of effects between *perceived control* and Big Five personality traits. The results indicate no clear directionality of influences. However, they show that decreases in *perceived control* are linked to changes in Big Five personality traits, in particular with increases in *neuroticism* and decreases in *conscientiousness*.

In sum, the results of our study provide strong support for different mean-level trends in old age in opposition to younger and middle aged adults that are probably due to different processes of adaptation in later adulthood. These findings are in line with the SOC model of successful aging (Baltes, 1997; Freund & Baltes, 1998). Thus, our study adds an important extension to the *maturity principle of personality development* as formulated by Roberts et al. (2008) for younger and middle aged adults with different implications for average trends and underlying developmental mechanisms in different ages.

Differential Stability in Old Age

Despite significant mean-level trends for specific personality traits, individual differences in personality traits seem to be largely stable over time in old age (supporting Hypothesis 3a). Consistent with our expectation (Hypothesis 4a), rank-order continuity in personality traits was attributable to both genetic and environmental influences. Compared to previous findings on the sources of personality development in young and middle adulthood, where genetic factors consistently represent the major source of continuity in personality differences (Bleidorn et al., 2009; Kandler et al., 2013), our study on older twins indicates generally lower genetic contributions and even larger influences of the environment on rank-order continuity in *neuroticism*, *agreeableness*, and *affect intensity*. This is in line with findings of increasing contributions of environmental factors to stabilize personality differences from younger to middle adulthood (Kandler et al., 2010; Viken et al., 1994) and suggests further accumulation of environmental contributions to differential personality stability in proportion to genetic influences until old age.

Accumulating environmental contributions to stability may reflect increasing stabilization of close relationships as driving forces of stability in old age. Consistent with this perspective, several studies have reported that the rank-order stability in structural features (e.g., network size and density) and qualitative aspects (e.g., reciprocity and support) of social

relationships increased from young to old adulthood, albeit a decrease in general network size and an increase in the depth of the remaining relationships with age (Asendorpf & Wilpers, 2004; Neyer & Lang, 2004; Lang, 2000; Lang, Staudinger, & Carstensen, 1998; Wrzus, Hänel, Wagner, & Neyer, 2013).

Stability in well-being was strongly induced by individual-specific environmental influences. The rather low and non-significant contribution of genetic influences to the variance in well-being appears to contradict previous research that revealed heritability estimates for well-being to be quite similar to the estimates for personality traits (see Bartels & Boomsma, 2009, for a review). However, most previous studies relied on general life satisfaction with the focus on life as a whole or trait-like measures of well-being (i.e., positive and negative affect) which, not surprisingly, showed substantial links to *neuroticism*, *extraversion*, and *conscientiousness* (Steel et al., 2008), and thus could be expected to show substantial heritability (Weiss, Bates, & Luciano, 2008). Our study implicates that individual differences in well-being related to rather momentary (e.g., for days, weeks, or months) life circumstances, mental states, and psychological health do not depend significantly on genetic influences.

Differential Plasticity in Old Age

Beyond average age trends and although the rank-ordering of individuals remained largely stable over time, older individuals significantly differed in their developmental trajectories in all personality traits and well-being. Since we used latent modeling analyses, individual differences in change were corrected for random fluctuation between measurement occasions due to error of measurement. That is, consistent with Hypothesis 3b, variation in trajectories between the two measurement occasions can be treated as true individual change and, thus, as true variation in plasticity in old age.

According to the *plasticity principle of personality development* (Roberts et al., 2008), individual differences in change should be due to environmental factors (Hypothesis 4b). The

results provided support for this hypothesis, highlighting environmental factors not shared by twins as significant source of individual differences in change for all personality traits (except for *extraversion*) and well-being. Moreover, nonshared environmental factors represented the primary source of change in *neuroticism*, *agreeableness*, *perceived control*, *affect intensity*, and well-being.

Experiences not shared by twins may reflect individual life events and circumstances. The probably most robust finding from studies that analyzed the interrelation between life events and personality change is the small prospective effect of extremely adverse events on *neuroticism* (e.g., Löckenhoff, Terracciano, Patriciu, Eaton, & Costa, 2009; Middeldorp, Cath, Vink, & Boomsma, 2008; Riese et al., 2014). Controlled for prospective personality effects, the occurrence of negative events appears to be prospectively linked to increases in *neuroticism* and decreases in *agreeableness* (Kandler, Bleidorn, Riemann, Angleitner, & Spinath, 2012). Given that the number of experienced stressful events increases in old age compared to middle adulthood (Plomin et al., 1990), their influence on personality may also increase. Individual differences in the number and intensity of important events experienced might explain the significant contributions of nonshared environmental influences on individual differences in change. In the first instance, thus, the findings support the *plasticity principle*.

A critical second view, however, offers some discrepant results: The genetic influences on individual differences in change were larger than environmental influences for *extraversion*, *openness*, and *conscientiousness* as well as comparable in size for *agreeableness* and *perceived control*. These results offer a new perspective on and extend existing knowledge about the sources of personality development in old age and throughout the life span: From childhood, adolescence, and young adulthood until the beginning of middle adulthood, genetically driven individual differences in change decrease up to the point where genetic factors appear to contribute exclusively to rank-order continuity (see Kandler, 2012; and Turkheimer et al., 2014).

In old age, however, it seems that genetic factors again drive individual differences in personality change. That is, the drop of rank-order continuity in personality in old age (Lucas & Donnellan, 2011; Wortman et al., 2012) is not only due to a decline in environmental continuity but also attributable to genetic sources.

Genetic sources of change may reflect individual differences in the genetically induced process of deterioration of cognitive, emotional, and social functionality and (or) genetically driven selection, optimization, and compensation mechanisms (e.g., selective physical and social activity to compensate physiological and cognitive losses). However, recent research suggests that physiological health declines have only small direct influences on changes in personality in old age (e.g., Berg & Johansson, 2014; Magee et al., 2013; Takahashi et al., 2013), which warrants more research on social, cognitive, and neurobiological factors and their interplay as predictors of personality change in older age (Klimstra, Bleidorn, Asendorpf, van Aken & Denissen, 2013). Our study thus extends this line of research by examining the interplay between personality traits and psychological health in old age.

The Interrelation between Personality and Well-Being in Old Age

Since recent studies have provided support for a reciprocal relationship between personality traits and psychological health (Soto, 2014; Specht, Egloff, & Schmukle, 2013b), we investigated whether change in well-being is linked to personality change in old age. However, we mentioned serious problems with trait-like measurements of well-being in previous studies which, not surprisingly, can be expected to show strong links to personality traits as well as estimates of heritability and rank-order stability comparable to those for personality trait measures. Thus, the use of such measures leads to biased results regarding the investigation of the interrelation between personality traits and well-being. We therefore examined the reciprocity of the interrelation between personality traits and well-being using a more state-sensitive measure of well-being.

In line with Hypothesis 5 and previous research (e.g., Boyce et al., 2013; Steel et al., 2008), we found that individual differences in levels and changes in well-being are significantly associated with individual differences in levels and changes in *neuroticism*, *extraversion*, *conscientiousness*, and *perceived control*. Contrary to Hypothesis 5 and previous findings, however, we did not find a significant interrelation between well-being and agreeableness. In addition, our findings do not implicate reciprocity of the effects between personality traits and well-being: Not in line with recent research (Soto, 2014; Specht et al., 2013b) but consistent with previous considerations (Diener et al., 1999; Headey, 2006; Kunzmann, Little, & Smith, 2002), we found prospective effects from personality traits (i.e., *neuroticism*, *conscientiousness*, and *perceived control*) on well-being but no significant well-being effects on personality traits (except for *perceived control* by trend). These prospective effects generally parallel level correlations as well as change correlations. That is, increases in *neuroticism* and decreases in *conscientiousness* and *perceived control* affect declines in well-being. Given the nonsignificant mean-level trend in well-being, the systematic interrelation with changing personality traits may be surprising. However, prospective trait effects and change correlations are generally lower compared to initial level correlations. That is, the larger the individual change in personality traits, the more likely it is linked to change in well-being.

Limitations and Future Directions

Using a two-wave longitudinal and genetically informative design with a sample of older adults, different personality measures, and latent model analyses to analyze patterns and sources of personality development in old age, our study exhibits a number of strengths. However, some limitations and avenues for future research have to be addressed.

The sample size of older twin pairs was relatively small for genetically informative twin analyses. There was also an overrepresentation of women relative to men and MZ relative to DZ twins in our sample. This limits the representativeness and the statistical power to detect

significant genetic influences in the presence of shared environmental influences and vice versa. Future studies need to replicate our results and, if they can analyze larger and more representative data sets, they can focus on more fine-grained effects and differences in the sources of development between men and women.

Even though our sample comprises adults aged 64 to 89 and thus goes beyond the age range that has been analyzed in most previous studies of personality development, having information that links personality development to mortality indicators, or using samples that include even older participants, may be informative for personality development in line with the *extended maturity principle*. Older adults or people who move closer to death may show different trends, in particular with respect to well-being (see Gerstorf et al., 2010).

Furthermore, the time interval of our longitudinal study was restricted to five years. This time frame might have been not long enough to find significant decreases in agreeableness and openness. Since our findings suggest decreases by trend, future studies with longer time frames may find significant trends and significant correlations between changes in *perceived control* and changes in *extraversion*, *openness*, and *agreeableness*.

The study of twins is limited to the treatment of genetic and environmental sources as unrelated factors. Though the investigation of relative contributions of genetic and environmental sources of continuity and change in individual differences over time and across different ages can yield important implications (see Kandler, 2012), genetic factors rather correlate and interact with environmental factors in complex ways. They affect behavior via underlying physiological and biochemical mechanisms, but not without environmental contribution. Thus, it is important to note that estimates of genetic contributions to continuity and change in individual differences may also reflect gene–environment correlation (e.g., niche-picking; Scarr & McCartney, 1983) and Gene × Environment interaction (i.e., interaction between genetic and environmental effects shared by twins; Purcell, 2002) to some degree.

Conversely, an interaction between genetic factors and environmental effects not shared by twins would inflate estimates of nonshared environmental effects, if those interactions are present but not taken into account. Thus, an increase of the contribution of nonshared environmental influences to personality stability may also reflect increases of stabilizing Gene × Environment interaction effects (see Bleidorn et al., 2014). Future studies may include measured environmental factors (e.g., normative and non-normative life events or social role transitions) into the examination of personality development in old age to more directly test this hypothesis and to provide deeper insight into the interplay between genetic and environmental factors in personality development across the life span.

Since neither psychological well-being (as our study suggests) nor physiological health declines (as recent studies suggest) substantially affect personality change, the question remains which mechanisms drive the reverse developmental trends in old age. Our study indicates that genetic influences may constitute one factor affecting change in some personality traits in old age. It is well known that genetically driven cognitive degeneration (i.e., dementia) is associated with personality changes (Rankin, Baldwin, Pace-Savitsky, Kramer, & Miller, 2005; Rubin, Morris, & Berg, 1987). Relatedly, Soto and colleagues (2011) reported dramatic drops in average trends of personality traits in the second decade of life at the typical time of puberty which reflects a genetically driven socio-biological transition. Subtle changes in biological degeneration may also account for personality change in old age. Consequently, it may play a role how health status is captured (i.e., subjective vs. objective or somatic vs. cognitive). Thus, future studies on personality development in old age may refer to this focus and should take individual differences in subtle but steady declines in health status into account.

Conclusions

This longitudinal study of older twins extends the knowledge of patterns and sources of personality development. Results suggest average personality change in old age that shows in

the opposite direction compared to the age trends typically reported for younger and middle aged adults (Roberts et al., 2006) implicating a refined formulation of the *maturity principle* of personality development for old age. Opposite trends in old age can be explained by different developmental and adaptive strategies (i.e., selection, optimization, and compensation; Baltes, 1997) compared to younger adults (e.g., increasing social integrity and identity formation; Roberts et al., 2008). In line with this perspective, change in *perceived control* was found to be associated with change in other personality traits reflecting adaptive mechanisms.

Even though individual differences in personality traits are largely stable over time, personality can change and individuals differ in change throughout the life span. Genetic as well as environmental sources affect both continuity in individual differences and variation in intraindividual change indicating that multiple sources - which may be correlated and interact in complex ways - drive personality development and differential plasticity in old age. Though our study suggests significant genetic contributions to change for some personality traits, individual environmental factors were the primary source of personality change in old age providing support for the *plasticity principle* of personality development (Roberts et al., 2008).

Continuity and change in well-being appear to be primarily due to environmental stability and fluctuation. Despite no significant normative change in well-being in old age, individual well-being can vary with change in personality traits. The nature of the interrelation between personality traits and well-being in old age appears to be primarily unidirectional indicating that personality change predicts change in well-being.

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Table 1. Reliability and rank-order continuity of personality trait and well-being measures

			Time 1		Time 2		Rank-order continuity		
		N		Split-		Split-		True	
Variables	Parcels	Items	α	half	α	half	Observed	score	Residuals
Neuroticism	1	6	.61		.55		.65		.31
	2	6	.63	.78	.60	.79	.62	.83	.25
Extraversion	1	6	.60		.57		.63		.43
	2	6	.58	.70	.57	.68	.64	.83	.46
Openness	1	6	.40		.40		.61		.28
	2	6	.48	.73	.57	.67	.56	.89	.18
Agreeableness	1	6	.45		.49		.62		.41
	2	6	.47	.67	.47	.65	.56	.84	.25
Conscientiousness	1	6	.67		.66		.64		.35
	2	6	.65	.80	.63	.79	.58	.81	.14
Perceived control	1	16	.73		.75		.66		.27
	2	16	.73	.83	.76	.86	.70	.82	.40
Affect intensity	1	20	.82		.84		.60		.39
	2	20	.81	.87	.82	.89	.61	.67	.45
Well-being	1	4	.60		.60		.29		-.12
	2	4	.66	.87	.61	.86	.34	.44	.07

Note. Reliability estimates were based on data from all participants who participated at Time 1 (*N* = 410) and Time 2 (*N* = 324). Rank-order continuity estimates (cross-time correlations) were based on data from all participants who participated at both measurement occasions (*N* = 324).

Table 2. Latent change model fitting results: Model parameter estimates

	Level				Change			
Variables	*M*	σ^2	r_{AGE}	r_{SEX}	*M*	σ^2	r_{AGE}	r_{SEX}
Neuroticism	**14.10**	**8.63**	.03	**-.29**	**0.38**	**2.93**	.15	.11
Extraversion	**19.59**	**6.47**	**-.13**	**.19**	**-0.41**	**2.07**	**-.19**	-.14
Openness	**20.48**	**7.72**	-.01	.10	-0.23	**1.63**	-.07	.16
Agreeableness	**23.99**	**3.89**	-.07	**-.35**	-0.13	**1.18**	.19	-.04
Conscientiousness	**24.65**	**6.71**	**-.22**	**.18**	**-0.36**	**2.64**	-.16	-.03
Perceived control	**64.44**	**48.28**	**-.15**	**.24**	**-1.12**	**17.59**	-.09	-.15
Affect intensity	**76.79**	**91.51**	.04	-.11	-0.17	**61.98**	-.06	-.08
Well-being	**2.88**	**1.10**	.01	**.21**	-0.04	**1.23**	-.10	-.05

Note. Parameter estimates are based on full information maximum likelihood model fitting results using all available data (N = 410 for Time 1 and N = 324 for time 2); r_{AGE} = correlations with age at time 1; r_{SEX} = correlations with sex (codes: 0 = female and 1 = male); significant estimates ($p < .05$) are shown in boldface.

Table 3. Latent-change twin model fitting results: Model parameter estimates

	Variance components					
	Level			Change		
Variables	a_L^2	c_L^2	e_L^2	a_C^2	c_C^2	e_C^2
Neuroticism	**3.86**	1.30	**3.46**	0.00	0.62	**2.08**
	45%	15%	40%	0%	23%	77%
Extraversion	**4.93**	0.00	**1.70**	**1.81**	0.00	0.05
	74%	0%	26%	97%	0%	3%
Openness	**4.88**	1.04	**1.75**	*0.78*	0.05	**0.50**
	64%	13%	23%	59%	4%	37%
Agreeableness	**1.19**	**1.72**	**0.86**	*0.40*	0.00	**0.56**
	32%	45%	23%	41%	0%	59%
Conscientiousness	**4.96**	0.31	**1.45**	**1.49**	0.00	**0.91**
	74%	4%	22%	62%	0%	38%
Perceived control	**39.60**	0.00	**8.09**	*6.69*	0.00	**8.29**
	83%	0%	17%	45%	0%	55%
Affect intensity	**37.45**	0.00	**53.85**	0.00	0.00	**52.82**
	41%	0%	59%	0%	0%	100%
Well-being	0.26	0.00	**0.84**	0.00	0.08	**0.79**
	23%	0%	77%	0%	9%	91%

Note. Parameter estimates are based on full information maximum likelihood model fitting results using all available data (complete 134 MZ and 63 DZ twin pairs at time 1 and complete 101 MZ and 47 DZ twin pairs at time 2); a^2 = genetic variance; c^2 = variance due to environmental influences shared by twins; e^2 = variance due to environmental influences not shared by twins; significant estimates ($p < .05$) are shown in boldface; significant genetic influences on change variance (a_C^2) after fixing $c_C^2 = 0$ are shown in italics.

Table 4. Crossed-lagged regression model fitting results for the interrelation between perceived control and Big Five personality traits: Model parameter estimates

	Initial	Continuity		Effects		Change
Big Five traits	correlation	β_T	β_{PC}	$\beta_{T\rightarrow PC}$	$\beta_{PC\rightarrow T}$	correlation
Neuroticism	**-.68**	**.91**	**.78**	-.07	.11	**-.47**
Extraversion	**.47**	**.84**	**.82**	.01	-.01	.12
Openness	**.29**	**.90**	**.81**	.06	-.01	*.18*
Agreeableness	**.19**	**.86**	**.83**	-.04	*-.12*	.16
Conscientiousness	**.54**	**.88**	**.77**	*.10*	**-.14**	**.41**

Note. Parameter estimates are based on full information maximum likelihood model fitting results using all available data (N = 410 for Time 1 and N = 324 for time 2); β_T and β_{PC}: rank-order continuity specific for Big Five traits and perceived control; prospective trait ($\beta_{T\rightarrow PC}$) and prospective control effects ($\beta_{PC\rightarrow T}$); significant model parameter estimates ($p < .05$) are shown in boldface; estimates in italics are significant by trend ($p < .10$).

Table 5. Crossed-lagged regression model fitting results for the interrelation between personality traits and well-being: Model parameter estimates

	Initial	Continuity		Effects		Change
Big Five traits	correlation	β_T	β_{WB}	$\beta_{T\rightarrow WB}$	$\beta_{WB\rightarrow T}$	correlation
Neuroticism	**-.47**	**.80**	**.32**	**-.25**	-.06	**-.21**
Extraversion	**.26**	**.83**	**.40**	*.13*	.01	.16
Openness	.04	**.89**	**.44**	.02	.04	-.16
Agreeableness	.04	**.84**	**.43**	.10	.00	-.09
Conscientiousness	**.27**	**.80**	**.38**	**.18**	.02	*.17*
Perceived control	**.36**	**.79**	**.38**	**.15**	*.09*	**.22**
Affect intensity	.01	**.67**	**.44**	.06	.06	.06

Note. Parameter estimates are based on full information maximum likelihood model fitting results using all available data (N = 410 for Time 1 and N = 324 for time 2); β_T and β_{WB}: rank-order continuity specific for personality traits and well-being; prospective trait ($\beta_{T\rightarrow WB}$) and prospective well-being effects ($\beta_{WB\rightarrow T}$); significant model parameter estimates ($p < .05$) are shown in boldface; estimates in italics are significant by trend ($p < .10$).

Figure 1. Latent change model: The means of parcels 1 are fixed to zero and the means of parcels 2 are constrained to be equal across measurement occasions in order to estimate average level and change scores. Individual differences in parcel scores for Time 2 are decomposed into variances due to level scores, change scores, two times the covariance between level and change scores, and measurement error (ε). The model additionally allows for correlations of level and change with age and sex.

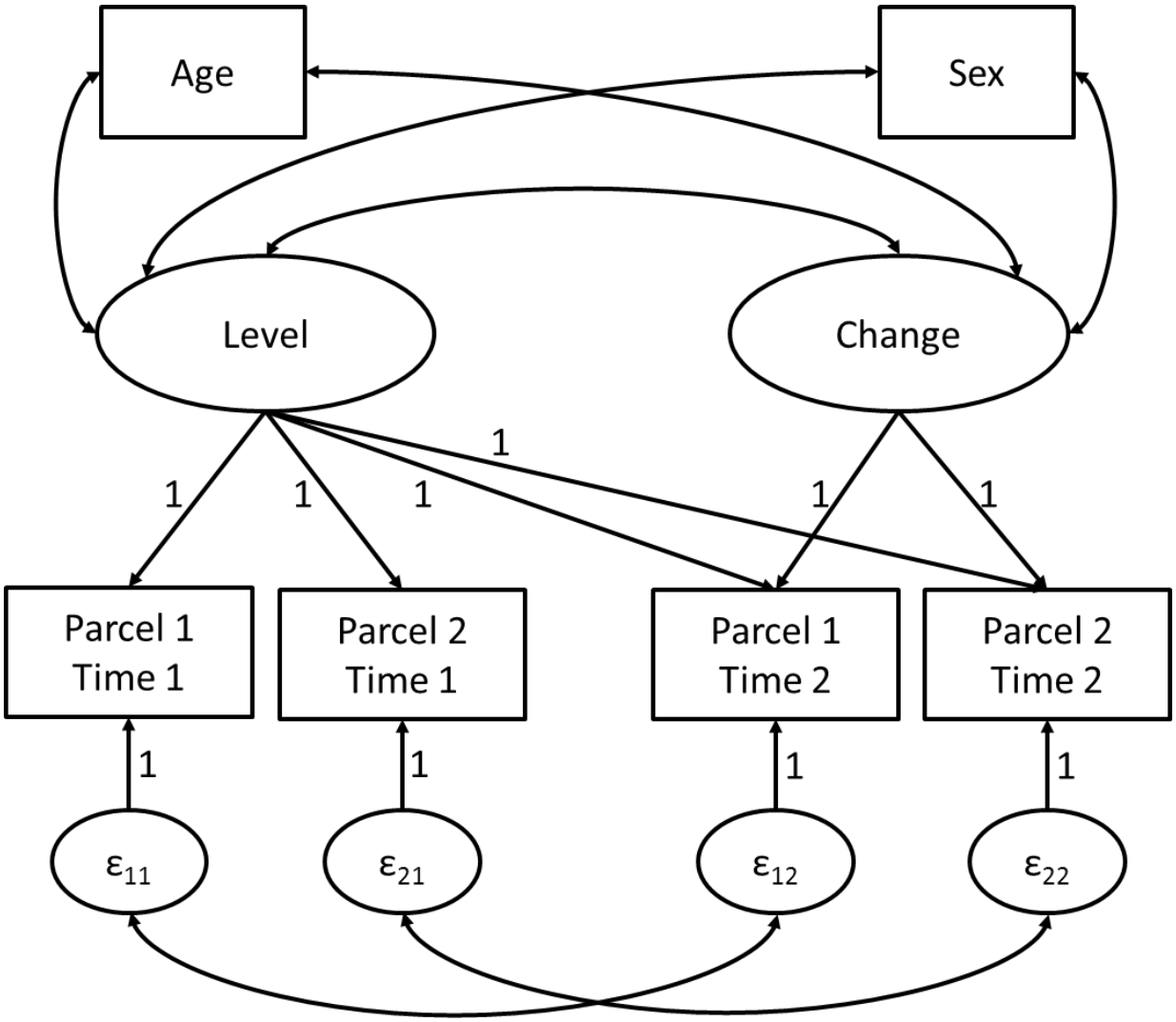

Figure 2. Latent change twin model: Controlled for variance due to random error (ε) and nonrandom parcel-specific method factors (M_1 and M_2), this model disentangles individual differences in level and change into variance components due to genetic influences (G_{LA} and G_{CA}; genetic correlations are σ = 1 for monozygotic twins and σ = .5 for dizygotic twins), environmental influences shared by twins (SE_{LA} and SE_{CA}), and environmental influences not shared by twins (NE_{LA} and NE_{CA}). All genetic and environmental variance components were fixed to one in order to estimate effects through path coefficients (a_L and a_C: additive genetic effects on level and change; c_L and c_C: shared environmental effects on level and change; e_L and e_C: nonshared environmental influences on level and change). See text for further model assumptions.

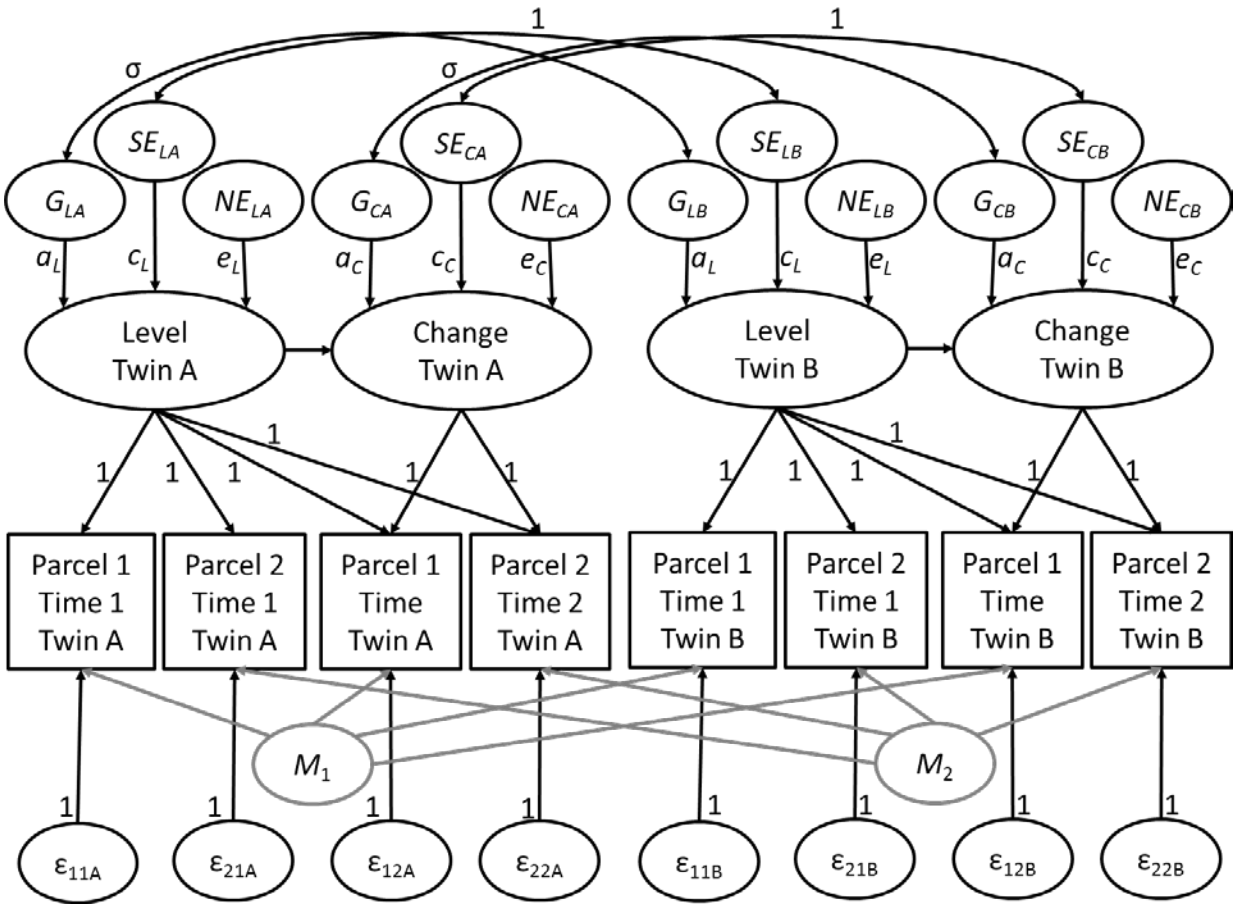

Figure 3. Latent crossed-lagged regression model: This figure illustrates the example model that investigates the interrelation between well-being (WB) and a trait variable. The model estimates initial correlations between variables (σ_I), variable-specific rank-order continuity (β_T and β_W), prospective trait ($\beta_{T\rightarrow W}$) and prospective well-being effects ($\beta_{W\rightarrow T}$), and change correlations (σ_C).

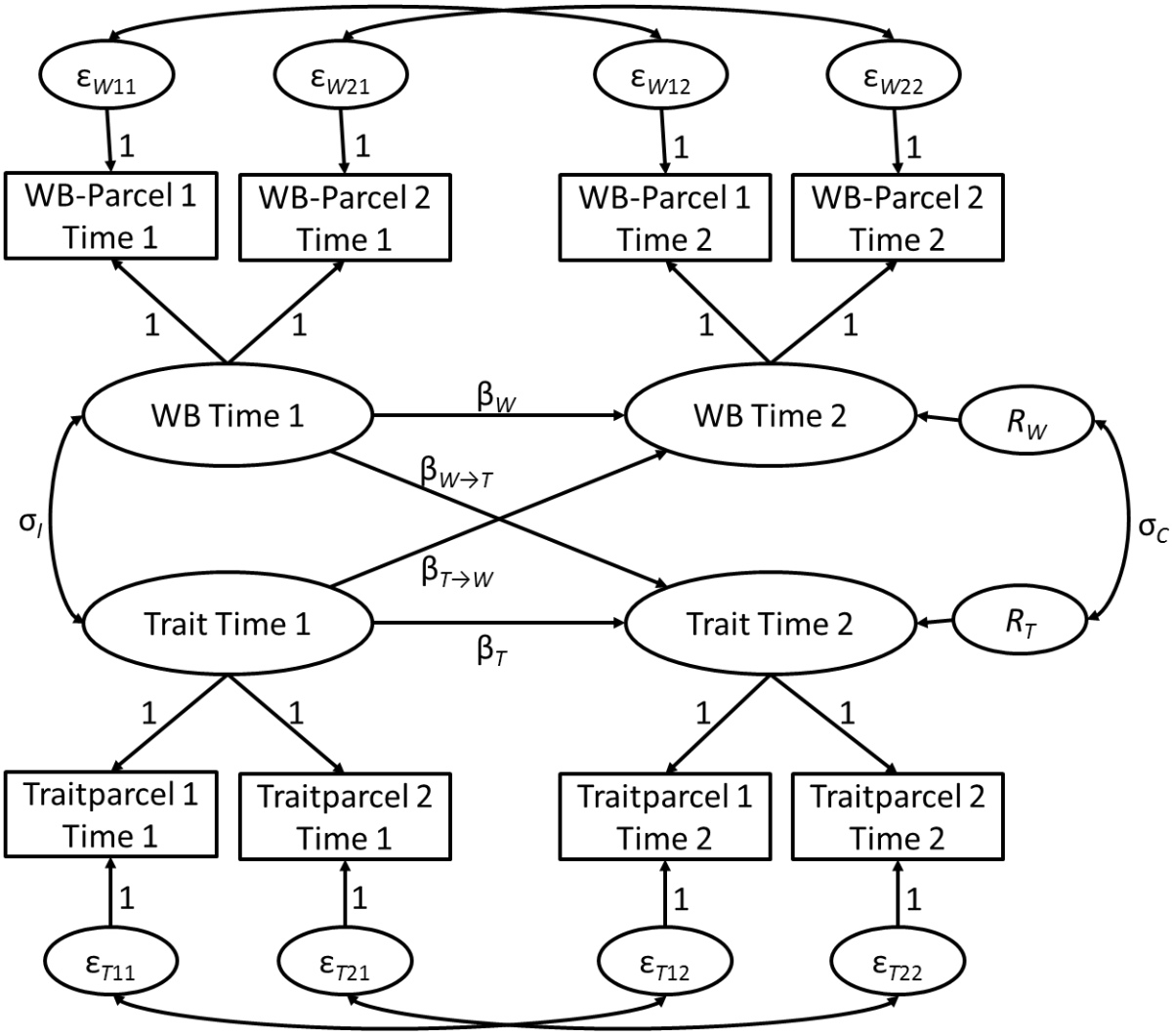

Figure 4. Z-standardized average change for all variables

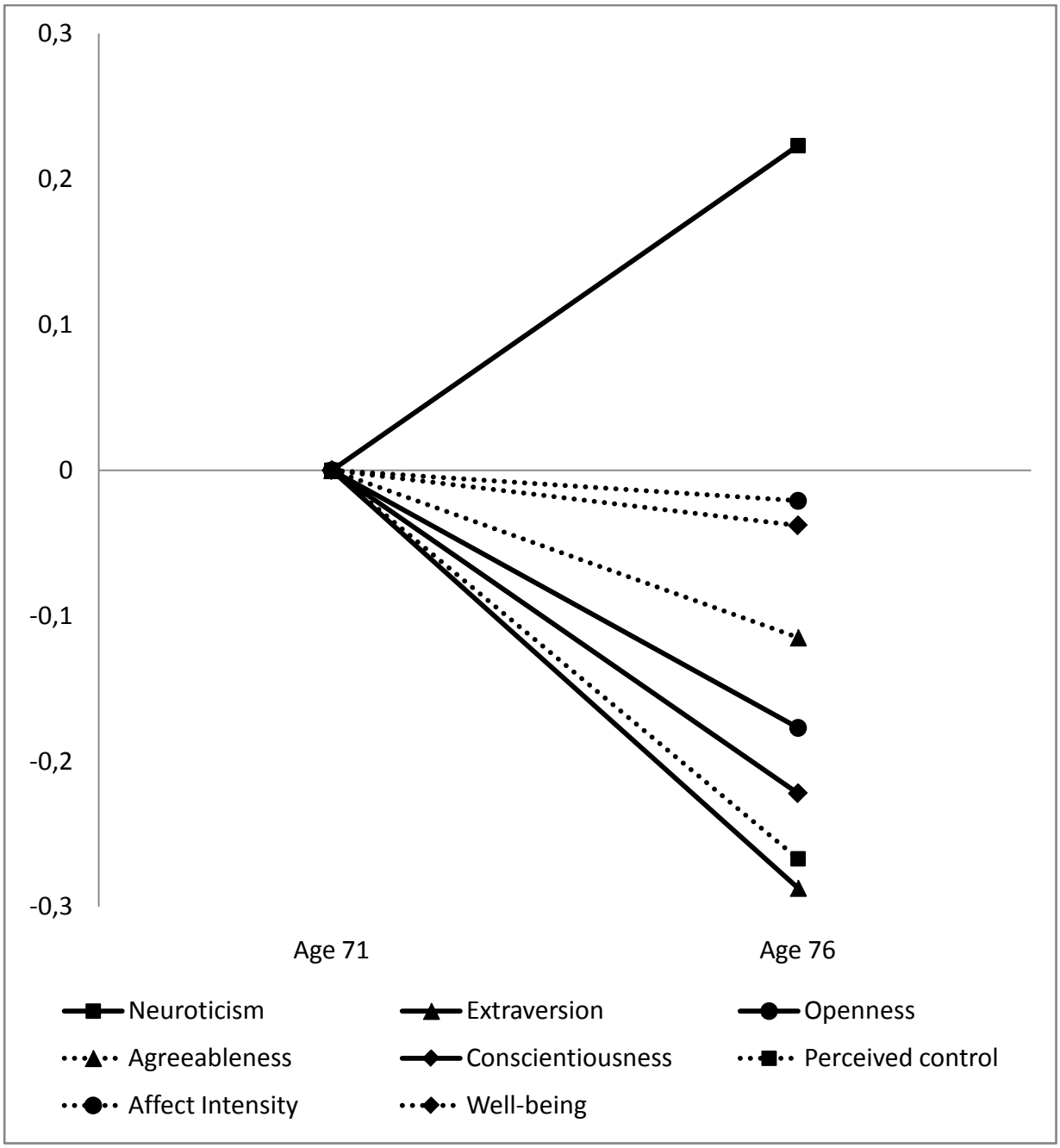

Table S1. Means and standard deviations of parcel scores for Time 1 and Time 2

			Time 1		Time 2	
Variables	Parcels	N_{Items}	*M*	*SD*	*M*	*SD*
Neuroticism	1	6	14.14	3.69	14.25	3.59
	2	6	13.36	3.69	13.60	3.68
Extraversion	1	6	19.55	3.44	19.35	3.38
	2	6	19.06	3.56	18.77	3.46
Openness	1	6	20.55	3.74	20.36	3.76
	2	6	20.18	3.60	20.38	3.83
Agreeableness	1	6	24.03	2.85	23.85	2.86
	2	6	24.17	2.67	24.13	2.74
Conscientiousness	1	6	24.66	3.14	24.27	3.32
	2	6	24.31	3.22	24.10	3.31
Perceived control	1	16	64.20	8.42	63.99	8.22
	2	16	64.05	8.27	62.90	8.07
Affect intensity	1	20	76.78	10.82	76.58	10.96
	2	20	78.79	10.87	78.49	10.85
Well-being	1	4	2.88	1.18	2.87	1.18
	2	4	2.92	1.23	2.89	1.20

Note. Statistics were based on data from all participants who participated at Time 1 (N = 410) and Time 2 (N = 324).

Table S2. True score correlations among variables within and between measurement occasions

		N		E		O		A		C		PC		AI	
Variables	Time	1	2	1	2	1	2	1	2	1	2	1	2	1	2
Extraversion	1	**-.49**	**-.38**												
	2	**-.39**	**-.53**												
Openness	1	**-.41**	**-.38**	**.43**	**.36**										
	2	**-.32**	**-.33**	**.41**	**.48**										
Agreeableness	1	**-.28**	-.18	.17	.13	.10	-.01								
	2	**-.20**	**-.22**	.11	.04	.01	-.04								
Conscientiousness	1	**-.41**	**-.37**	**.45**	**.35**	**.18**	.09	**.42**	.21						
	2	**-.28**	**-.49**	**.26**	**.46**	.00	.06	**.23**	**.47**						
Perceived control	1	**-.68**	**-.51**	**.48**	**.39**	**.29**	**.25**	**.18**	.04	**.53**	**.32**				
	2	**-.59**	**-.60**	**.38**	**.34**	**.29**	**.29**	.12	.05	**.51**	**.48**				
Affect intensity	1	.14	.15	**.22**	**.20**	.07	.05	**.23**	**.30**	**.28**	**.19**	-.02	-.04		
	2	.02	-.02	**.31**	**.25**	.10	.09	**.26**	**.36**	**.31**	**.28**	.02	.03		
Well-being	1	**-.47**	**-.45**	**.28**	**.24**	.03	.08	.04	.04	**.27**	**.24**	**.35**	**.37**	.01	.06
	2	**-.38**	**-.44**	**.25**	**.29**	.04	-.03	.12	.07	**.27**	**.33**	**.28**	**.38**	.06	.11

Note. Correlation estimates were based on data from all participants who participated at Time 1 (N = 410) and Time 2 (N = 324); significant correlations (p < .01) are shown in boldface.

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